Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 18

Functions, Range, and Composition

Covers Chartrand & Zhang, Section 5.3

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Check both failure modes

Is it a function?

A relation $g \subseteq A \times B$ fails to be a function from A to B if

1. some element of A has **no** image, or
2. some element of A has **more than one**.

Check both. Repeated values in B are fine — only the A side is constrained.

Three words, kept straight

Domain = where inputs come from. **Codomain** = the declared target set. **Range** = what actually gets hit.
Range $\subseteq$ codomain always, and equality is a special condition with its own name ("onto," on Wednesday).

Three mistakes to expect

1. **Confusing range with codomain.** The codomain is declared; the range is computed.
2. **Reading $g \circ f$ left to right.** f is applied first.
3. **Assuming $f \circ g = g \circ f$.** They are usually different.

Part II: Practice Problems

Section A — Is it a function?

1. Let $A = \{1, 2, 3, 4\}$ and $B = \{p, q, r\}$. For each relation, decide whether it is a function from A to B, with a reason.

(a) $\{(1, p), (2, q), (3, p), (4, r)\}$

(b) $\{(1, p), (2, q), (3, r)\}$

(c) $\{(1, p), (1, q), (2, r), (3, p), (4, q)\}$

(d) $\{(1, q), (2, q), (3, q), (4, q)\}$

Section B — Domain, codomain, range

2. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c, d\}$, and $f = \{(1, b), (2, d), (3, b), (4, a), (5, d)\}$.

(a) State the domain, codomain, and range of f.

(b) Compute $f(\{1, 3\})$ and $f(\{2, 4\})$.

(c) Is there a subset $X \subseteq A$ with $|X| = 2$ and $|f(X)| = 1$? Give one or explain why not.

Section C — Composition

3. Let $A = \{1, 2, 3\}$, $B = \{w, x, y\}$, $C = \{5, 6, 7\}$, with

$$f = \{(1, x), (2, w), (3, x)\}, \qquad g = \{(w, 6), (x, 5), (y, 7)\}.$$

Compute $g \circ f$ as a set of ordered pairs, and give its range.

4. Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x - 4$ and $g(x) = x^2$.

(a) Find formulas for $g \circ f$ and $f \circ g$.

(b) Evaluate both at $x = 1$ and at $x = 4$.

(c) Are there any values of x where the two agree? Find them.

Section D — Writing

5. In five to eight complete sentences, explain the difference between the codomain and the range of a function, and why the codomain must be stated rather than computed. Give a concrete example where two functions share a rule and a domain but differ in codomain.

Section E — Challenge (optional)

6. Let A and B be finite sets with $|A| = m$ and $|B| = n$. How many functions are there from A to B? Justify.